

CSE400 – Fundamentals of Probability in Computing

Lecture 19: N-Variate Joint PDF / PMF / CDF

Topic: Joint Distribution of N Random Variables

1. Joint Distribution of N Random Variables

1.1 Definition

Let $X_1, X_2, \dots, X_n$ be n random variables defined on a common probability space.

The joint cumulative distribution function (CDF) is defined as:

$$F_{X_1, \dots, X_n}(x_1, \dots, x_n) = P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n)$$

Step-by-Step Reasoning

1. Consider each random variable individually.
2. Form the joint event:

$$(X_1 \leq x_1) \cap (X_2 \leq x_2) \cap \dots \cap (X_n \leq x_n)$$

3. Compute the probability of this intersection.
4. This defines the joint CDF.

2. Example 1

2.1 Reasoning Procedure

1. Identify all random variables.
2. Construct the joint outcome space.
3. Assign PMF (discrete) or PDF (continuous).
4. Verify normalization:

Discrete case:

$$\sum_{x_1} \dots \sum_{x_n} p(x_1, \dots, x_n) = 1$$

Continuous case:

$$\int f(x_1, \dots, x_n) dx_1 \dots dx_n = 1$$

3. Application: Large MIMO System

3.1 Interpretation

Multiple antennas correspond to multiple random variables, forming an n -dimensional random vector.

3.2 Step-by-Step Reasoning

- Let each antenna output be X_i .
- Form the vector $(X_1, X_2, \dots, X_n)$.
- Describe system behavior using joint distribution.
- Capture dependencies among variables.

4. Joint PMF, CDF, PDF

4.1 Joint PMF (Discrete Case)

$$p(x_1, \dots, x_n) = P(X_1 = x_1, \dots, X_n = x_n)$$

Reasoning

- Each tuple represents an outcome.
- Assign probability to each tuple.
- Ensure normalization:

$$\sum_{x_1} \cdots \sum_{x_n} p(x_1, \dots, x_n) = 1$$

4.2 Joint CDF

$$F(x_1, \dots, x_n) = P(X_1 \leq x_1, \dots, X_n \leq x_n)$$

Reasoning

- Convert point probabilities to cumulative form.
- Evaluate joint inequalities.
- Function is non-decreasing in each argument.

4.3 Joint PDF (Continuous Case)

$$f(x_1, \dots, x_n) = \frac{\partial^n}{\partial x_1 \cdots \partial x_n} F(x_1, \dots, x_n)$$

Reasoning

- Start with joint CDF.
- Differentiate with respect to all variables.
- Obtain joint density.

4.4 Probability Computation

$$P(a_i \leq X_i \leq b_i) = \int_{a_1}^{b_1} \cdots \int_{a_n}^{b_n} f(x_1, \dots, x_n) dx_1 \cdots dx_n$$

5. Independence of Random Variables

5.1 Definition

Random variables $X_1, \dots, X_n$ are independent if:

$$p(x_1, \dots, x_n) = \prod_{i=1}^n p_{X_i}(x_i)$$

or

$$f(x_1, \dots, x_n) = \prod_{i=1}^n f_{X_i}(x_i)$$

Step-by-Step Reasoning

1. Compute joint distribution.
2. Compute marginal distributions.
3. Multiply marginals.
4. Compare:
 - Equal $\Rightarrow$ independent
 - Not equal $\Rightarrow$ dependent

6. Example 2: Uniformly Distributed 3D Vector

6.1 Setup

Let (X, Y, Z) be uniformly distributed in a region.

6.2 Joint PDF

$$f(x, y, z) = \begin{cases} c, & \text{inside region} \\ 0, & \text{outside} \end{cases}$$

6.3 Derivation

- Uniform $\Rightarrow$ constant density c
- Apply normalization:

$$\int_{\text{region}} c \, dV = 1$$
$$c = \frac{1}{\text{Volume of region}}$$

7. Example 3: Trivariate Random Variables

7.1 Joint CDF

$$F(x, y, z) = P(X \leq x, Y \leq y, Z \leq z)$$

7.2 Joint PDF

$$f(x, y, z) = \frac{\partial^3}{\partial x \partial y \partial z} F(x, y, z)$$

7.3 Marginal Distributions

$$f_X(x) = \int \int f(x, y, z) \, dy \, dz$$

$$f_Y(y) = \int \int f(x, y, z) \, dx \, dz$$

$$f_Z(z) = \int \int f(x, y, z) \, dx \, dy$$

Reasoning

- Start with joint PDF.
- Integrate out unwanted variables.
- Obtain marginals.

8. Logical Structure of the Lecture

- Define joint distribution
- Introduce PMF, CDF, PDF
- Study independence

- Apply to engineering systems (MIMO)
- Solve examples:
 - General joint case
 - Uniform 3D vector
 - Trivariate variables

9. Exam-Focused Key Points

- Joint CDF is the foundational definition
- PMF and PDF derive from it
- Independence $\Rightarrow$ factorization into marginals
- Marginals obtained via integration/summation
- Uniform distributions require normalization